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| **Status** | Finished |
| **Started** | Thursday, 8 October 2026, 8:16 PM |
| **Completed** | Thursday, 8 October 2026, 8:20 PM |
| **Duration** | 4 mins 8 secs |
| **Grade** | **15.00** out of 15.00 (**100**%) |

Question **1**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It uses the heap data structure
b. It bubbles up the largest element in each pass
c. It recursively sorts subproblems
d. It divides the list into sublists

The correct answer is: It bubbles up the largest element in each pass

Question **2**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Backtracking
b. Divide and conquer
c. Greedy method
d. Dynamic programming

The correct answer is:
Divide and conquer

Question **3**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys

b. A sorting algorithm is stable if it preserves the order of duplicate keys

c. A sorting algorithm is stable if it preserves the order of non-duplicate keys

d. A sorting algorithm is stable if it preserves the order of all keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **4**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Complexity

b. Sorting

c. Merging

d. Searching

The correct answer is:
Complexity

Question **5**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. sort()

b. order()

c. arrange()

d. sorted()

The correct answer is: sort()

Question **6**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Merge Sort
- b. Linear Search
- c. Quick Sort
- d. Bubble Sort

The correct answer is: Merge Sort

Question **7**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Linear Search
- b. Binary Search
- c. Merge Sort
- d. Bubble Sort

The correct answer is: Bubble Sort

Question **8**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the largest element to its correct position
- b. Bubbling up the smallest element
- c. Sorting the entire list in one pass
- d. Dividing the list into halves

The correct answer is: Bubbling up the largest element to its correct position

Question **9**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It makes the data unsorted
b. It complicates the selection of items
c. It makes it easier to select items based on their relationship to the rest of the items
d. It slows down the process

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **10**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Extraction
b. Selection
c. Distribution
d. Insertion

The correct answer is:
Insertion

Question **11**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

a. Finding a specific element in a list
b. Arranging data in a specific order
c. Removing duplicates from a list
d. Merging two datasets

The correct answer is: Arranging data in a specific order

Question **12**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Heap sort
b. Selection sort
c. Quick sort
d. Merge sort

The correct answer is:
Merge sort

Question **13**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Bubble Sort
b. Merge Sort
c. Linear Search
d. Binary Search

The correct answer is: Merge Sort

Question **14**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Being the most efficient sorting algorithm
b. Bubbling up the largest element to its correct position with each pass
c. Using the divide-and-conquer approach
d. Sorting data in a non-sequential manner

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **15**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 200,470,80,150,40,90,300,400,70,120

b. 40,80,90,150,200,300,400,470,70,120

c. 80,150,200,470,40,90,300,400,70,120

d. 40,70,80,90,120,150,200,300,400,470

The correct answer is:
80,150,200,470,40,90,300,400,70,120